Difference-in-Differences - 13 Feb 2025

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Statistical Study Group

Learning Objects

1. **Review: Why endogenous Matter?**
2. **Difference-in-Differences (DiD)**
3. **Empirical setting – shock setting**
4. **Validity Conditions**
5. **Real example: minimum wage law (Card and Kruger, 1994, AER)**

1. Why endogenous Matter?

1. Endogenous

Violations of the condition E[e|x]=0

1. **Omitted variable bias**
2. Reverse causality
3. Measurement errors
4. Selection bias

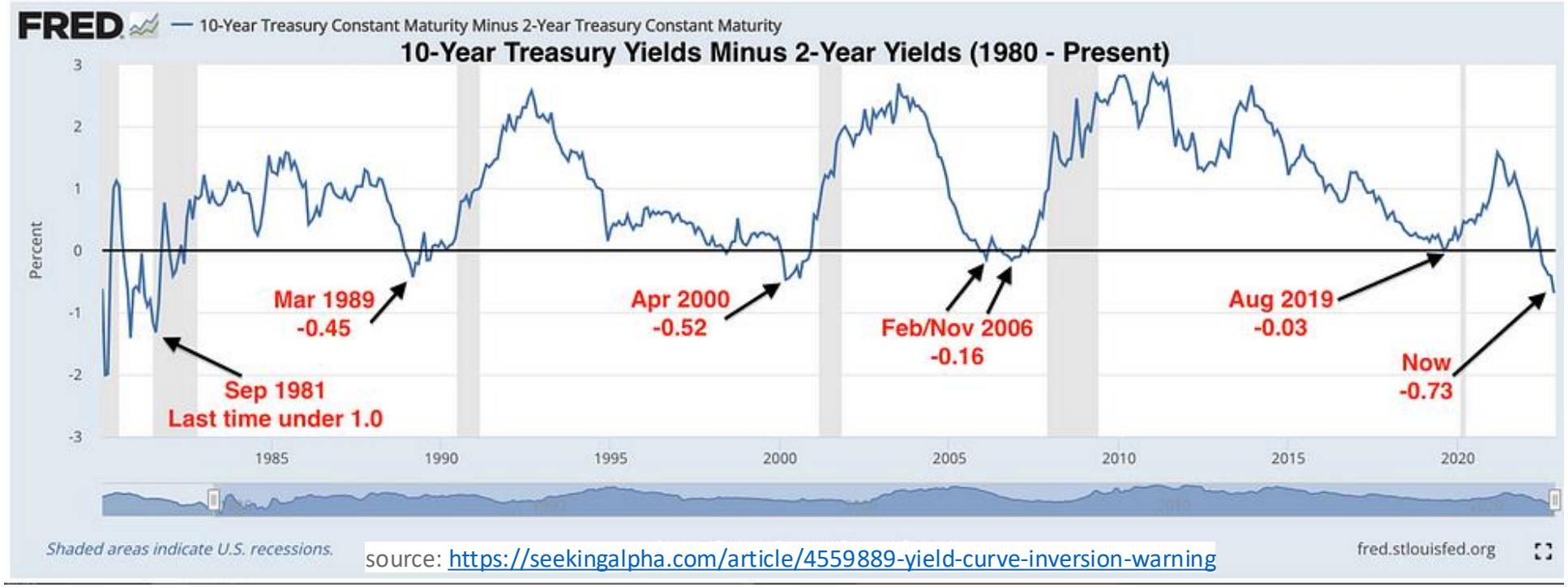
$$y = \beta_0 + \boldsymbol{\beta_1} x_1 + \beta_2 x_2 + \varepsilon$$

$$x_2 = \gamma_0 + \gamma_1 x_1 + \epsilon$$

$$y = \theta_0 + \boldsymbol{\theta_1} x_1 + e$$

$$\boldsymbol{\theta_1} = \boldsymbol{\beta_1} \, ?$$

1. Endogenous



source: https://seekingalpha.com/article/4559889-yield-curve-inversion-warning

Inverted yield curve => future economic recession; possible Granger Causality

- Causal? Inverted yield curve cause a future recession?
- Too early:
  1. Investors foresee future recession => buy long bond > short => - long bond yield
  2. Other reasons cause an inverted yield curve and later recession. (FED monetary policy)

1. Endogenous

Solution:

If you only care about forecasting, not causal interpretation => **No actions** are needed.

Casual: Looking for exogenous variation in the exogenous variable of interest (X)

1. Add x_2 in the regression
2. *Instrumental variables (IVs)*
3. Fixes effect models (not enough)
4. **Difference-in-difference estimates**
5. Research discontinuity design

2. Difference-in-differences estimates

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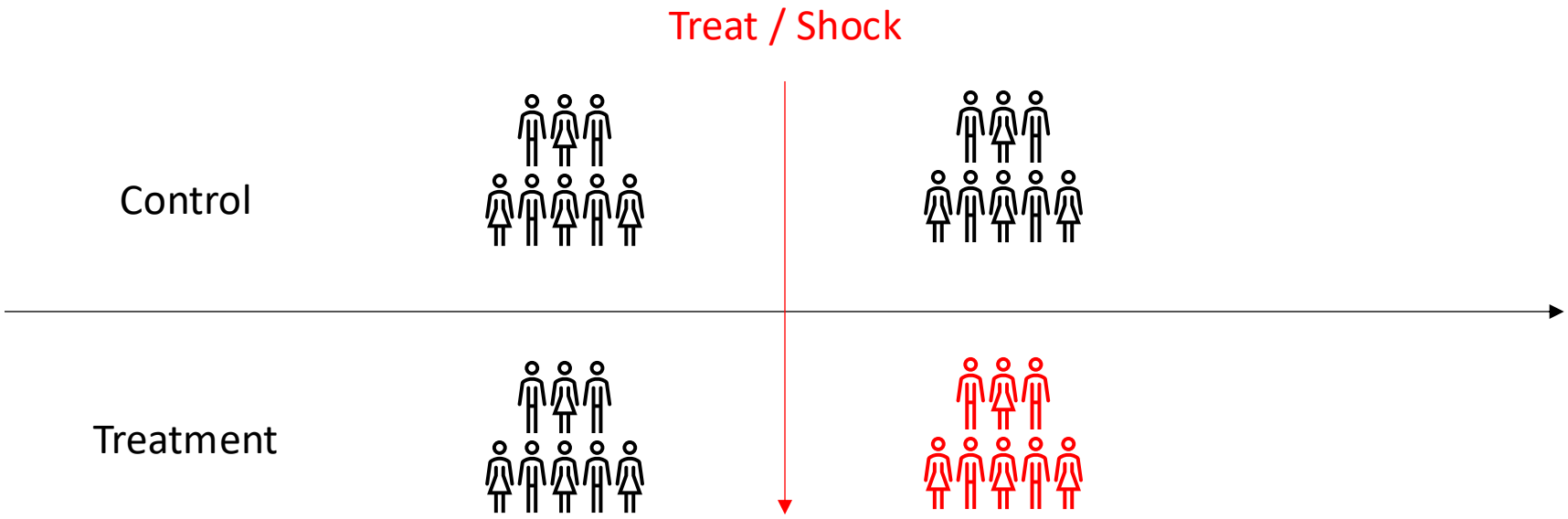

$$DiD = (Y_{Treat,Post} - Y_{Treat,Pre}) - (Y_{Control,Post} - Y_{Control,Pre})$$

3. Empirical setting – shock setting

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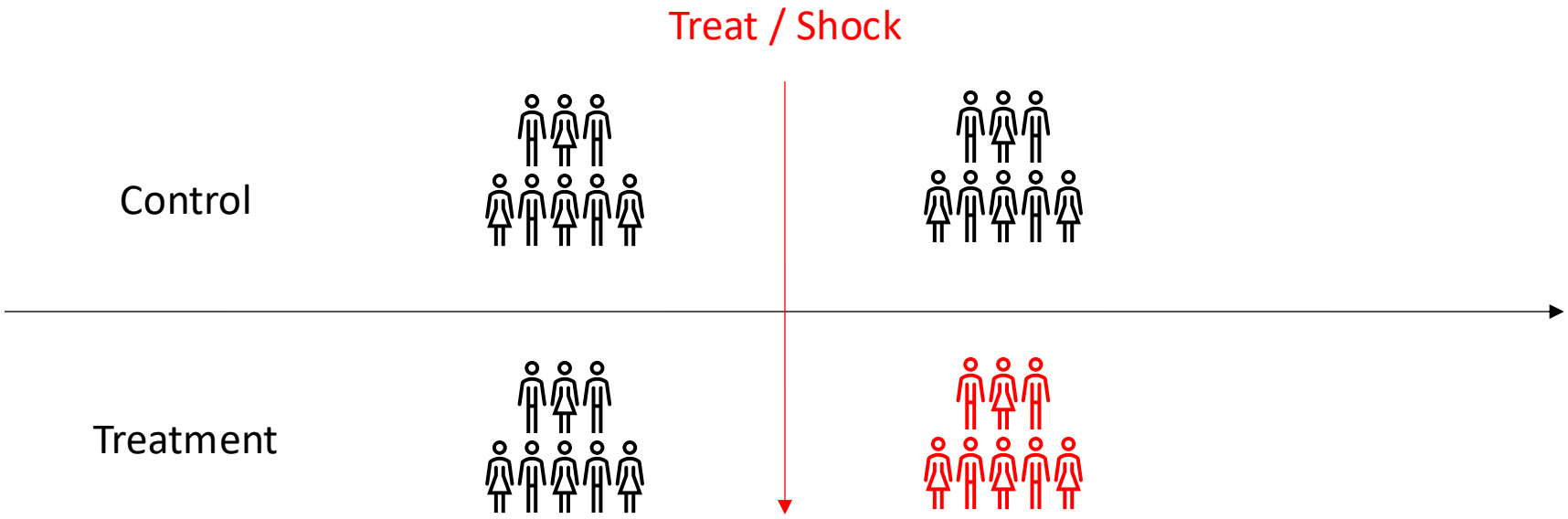

$$y = \beta_0 + \beta_1 Post + \beta_2 Treated + \beta_3 Post * Treated + Controls + FEs$$

Group\Time	Before	After	Diff (A-B)
Control	β_0	$\beta_0 + \beta_1$	β_1
Treatment	$\beta_0 + \beta_2$	$\beta_0 + \beta_1 + \beta_2 + \beta_3$	$\beta_1 + \beta_3$
Diff (T-C)	β_2	$\beta_2 + \beta_3$	β_3 (DiD)

4. Validity Conditions

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Challenges of the DiD:

- The shock should be **Exogenous**; policy, natural disasters, etc.
- Treated and Control groups should be **similar** enough in the outcome (e.g., Y)
  - Control best be Treated without the shock.

5. Real examples

5. Real examples: Card and Krueger (1994, AER)

Card and Krueger (1994, AER):

- Shock: The 1992 New Jersey Minimum Wage Experiment;
- New Jersey state: mini wage + from $4.25 to $5.05
- Pennsylvania state: unchanged mini wage
- Fast food restaurants in both states
- Treated * Post (X) => Employment (Y_{ist}), restaurant i, state s, time t.

5. Real examples: Card and Krueger (1994, AER)

Variable	Stores by state		
	PA (i)	NJ (ii)	Difference, NJ – PA (iii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	–2.89 (1.44)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	–0.14 (1.07)
3. Change in mean FTE employment	–2.16 (1.25)	0.59 (0.54)	2.76 (1.36)

$$\text{DiD} = (Y_{Treat,Post} - Y_{Treat,Pre}) - (Y_{Control,Post} - Y_{Control,Pre})$$

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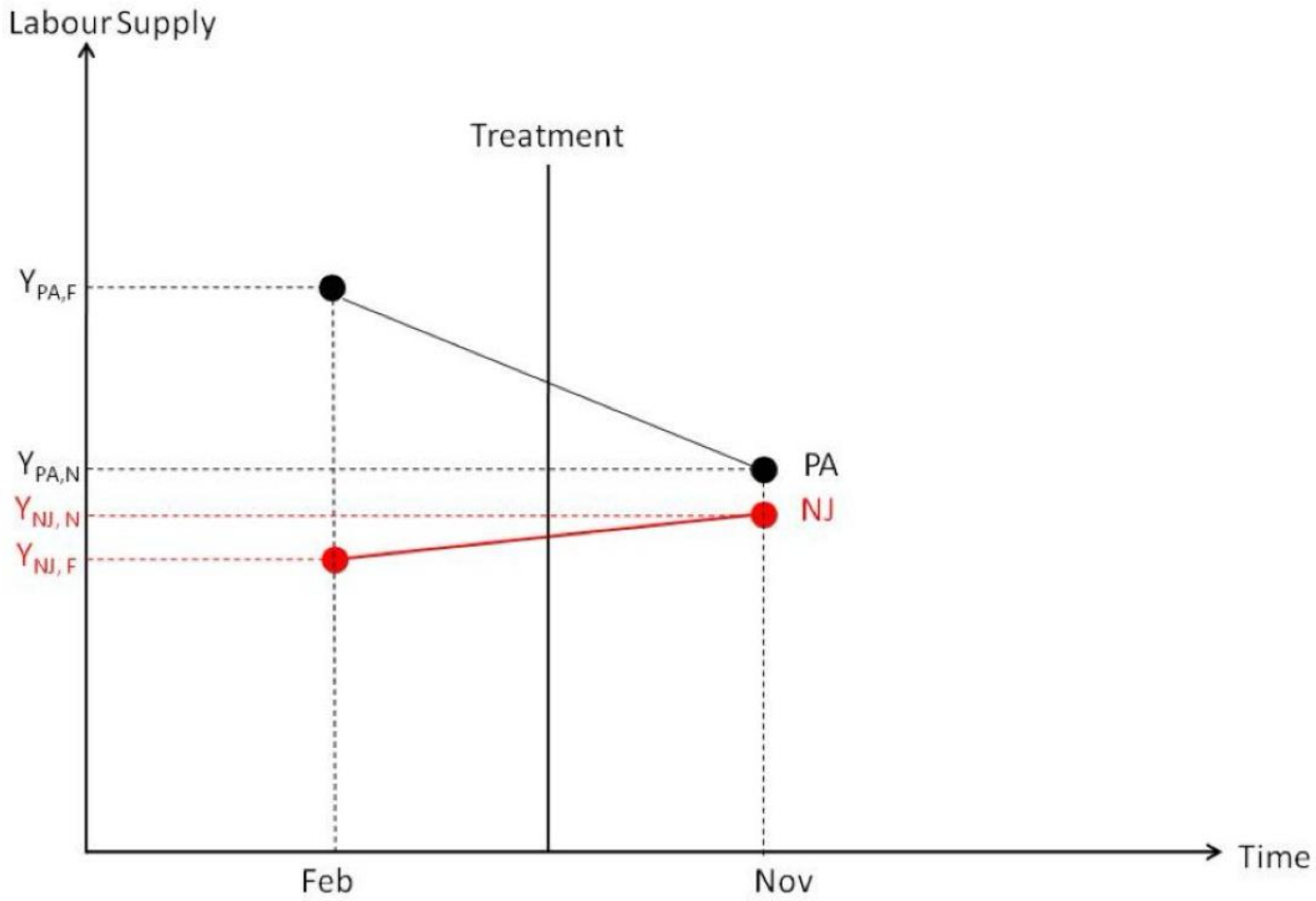


5. Real examples: Card and Krueger (1994, AER)

$$y = \beta_0 + \beta_1 Post + \beta_2 Treated + \beta_3 Post * Treated$$

$$y = \beta_0 + \beta_1 Post + \beta_2 NJ + \beta_3 Post * NJ$$

$$\Delta y = \beta_0 + \beta_1 NJ + Controls$$

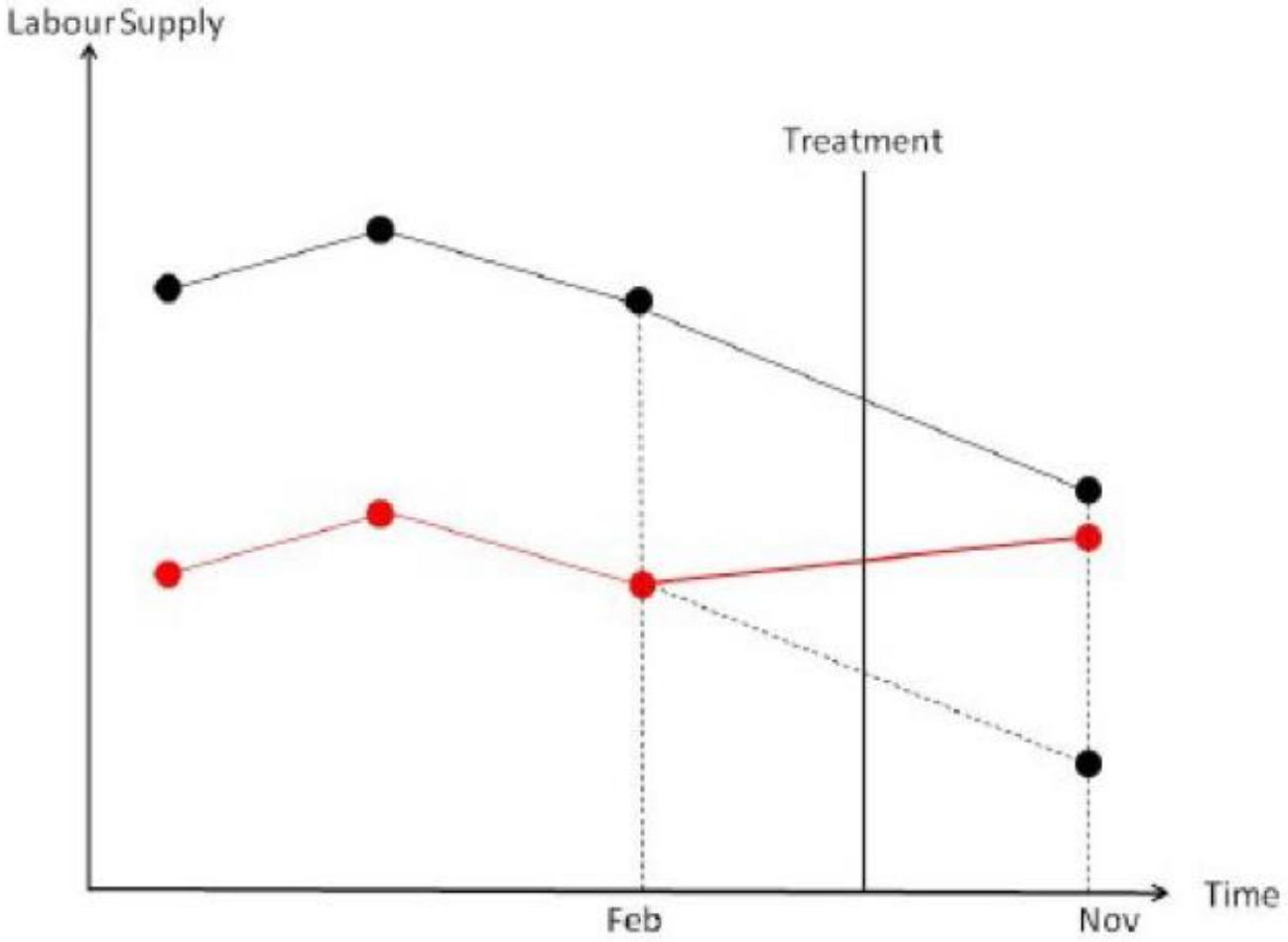
Δy : Employment change after the shock time in either group.

Independent variable	(i)	(ii)
1. New Jersey dummy	2.33 (1.19)	2.30 (1.20)
2. Initial wage gap[a]	—	—
3. Controls for chain and ownership[b]	no	yes
4. Controls for region[c]	no	no
5. Standard error of regression	8.79	8.78
6. Probability value for controls[d]	—	0.34

5. Real examples: Check the similarity of group

Check the similarity of groups:

- **Wish**: Treated and control groups have similar patterns in Y **without** the Shock **after-shock**.
- **Check**: Treated and control groups have similar patterns in Y **before-shock**.



Summary

1. **Endogenous**
   - => biased coefficients estimations
   - Can be used to predict but ≠ casual interpretations
2. **DiD:**
   - 1) Exogenous
   - 2) similarity
3. **Regression results:**
   - $y = \beta_0 + \beta_1 Post + \beta_2 Treated + \beta_3 Post * Treated + Controls + FEs$
   - Parallel trends

1. Endogenous: Omitted variable bias

$$y = \theta_0 + \theta_1 x_1 + e \quad \Rightarrow$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

$$= \beta_0 + \beta_1 x_1 + e$$

$$E(y|x_1) = \beta_0 + \beta_1 x_1 + E(e|x_1)$$

$$E(e|x_1) = E\left(\beta_2 x_2 + \varepsilon|x_1\right) = \beta_2 E(x_2|x_1) + E(\varepsilon|x_1)$$

$$E(x_2|x_1) = E(\gamma_0 + \gamma_1 x_1) = \gamma_0 + \gamma_1 x_1$$

$$\Rightarrow E(e|x_1) = \beta_2(\gamma_0 + \gamma_1 x_1)$$

$$E(y|x_1) = \beta_0 + \beta_1 x_1 + \beta_2\gamma_0 + \beta_2\gamma_1 x_1$$

$$= \beta_0 + \beta_2\gamma_0 + (\beta_1 + \beta_2\gamma_1)x_1$$

$$\theta_0 = \beta_0 + \beta_2\gamma_0$$

$$\theta_1 = \beta_1 + \beta_2\gamma_1$$

$$\theta_1 - \beta_1 = \beta_2\gamma_1$$

If $\beta_2\gamma_1 > 0 : \theta_1$ **over estimates** β_1

If $\beta_2\gamma_1 < 0 : \theta_1$ **under estimates** β_1

6. Real examples: (Goldstein et al., 2023, JAR)

5. Real examples: Goldstein et al. (2023, JAR)

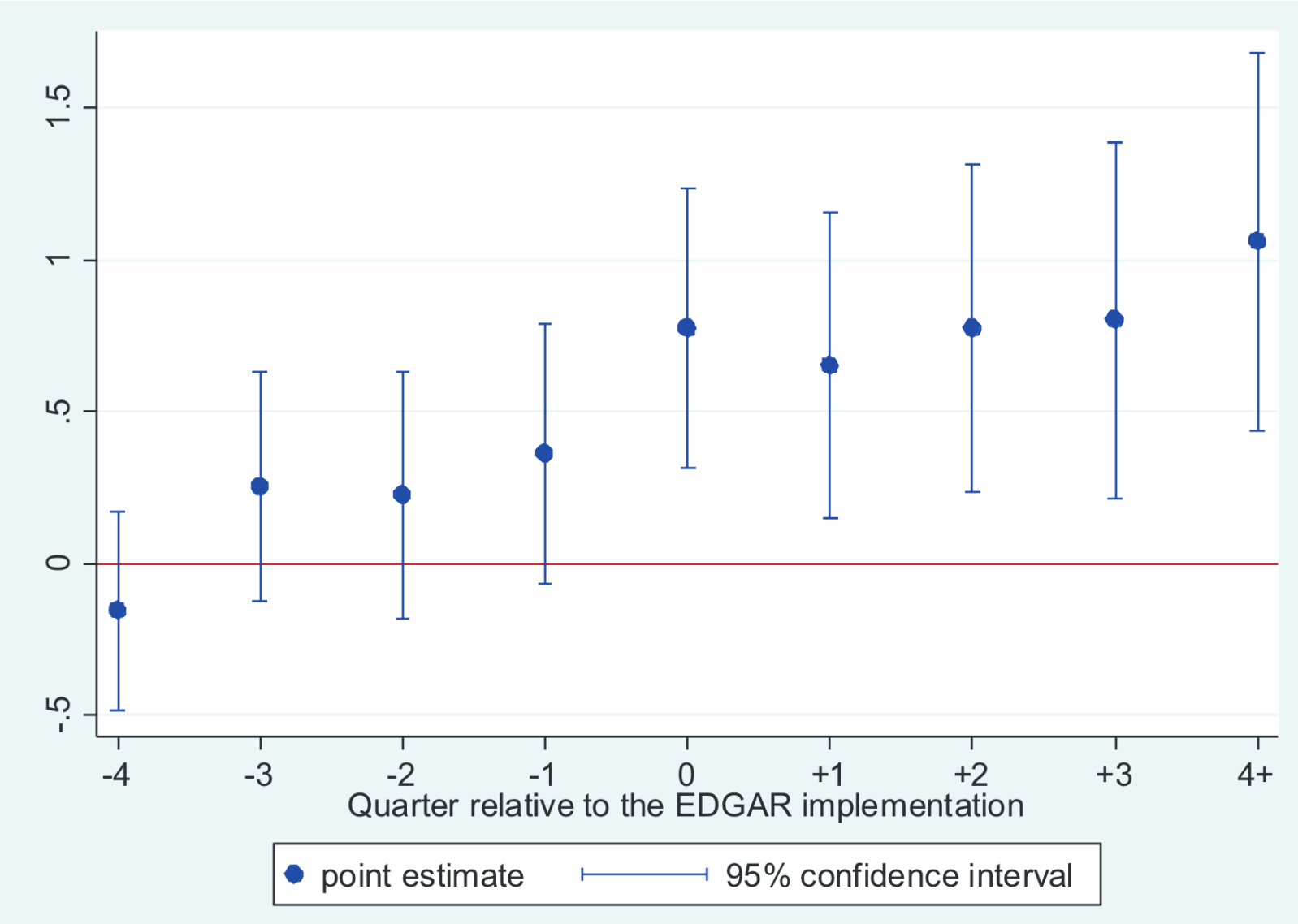
Goldstein et al. (2023, JAR): Staggard DiD

- Shock: mandatory transfer from paper fillings to electronic fillings, SEC-registered firms
- Hard-to-access paper copies to easy-to-access online fillings => - cost of info acquisition
- **Mandatory 10 phases**: not the same time
- 1993Q2 to 1996Q2, fillings available for the 1st phase corporates (CF-01)
- EDGAR (X) => Firm investment (Y), CAPEX / PPE

5. Real examples: Goldstein et al. (2023, JAR)

$$y = \beta_0 + \beta_1 \, EDGAR + Controls + FEs$$

Panel A: Baseline analysis		
Dependent variable =	*INVESTMENT*	
	(1)	(2)
EDGAR	**0.613*** **	**0.403*** **
	(4.05)	**(2.84)**
Q		1.714***
		(18.97)
CF		0.178***
		(12.94)
SIZE		0.354**
		(2.10)
Firm FE	Yes	Yes
Year-quarter FE	Yes	Yes
Observations	66,628	66,628
Adjusted R^2	0.272	0.302

5. Real examples: Goldstein et al. (2023, JAR)

Parallel trend: whether treated and control groups have similar investment trend before the shock.